

6(a)	any four from: 1 breed / cross, plants with desired (named) characteristics ; 2 breed, better / best, offspring, repeatedly / for many generations ; 3 (correct ref. to) <u>inbreeding</u> / <u>line breeding</u> ; 4 (correct ref. to) <u>hybridisation</u> / <u>outbreeding</u> / <u>outcrossing</u> ; 5 (gives) <u>hybrid vigour</u> / <u>heterosis</u> / <u>outbreeding enhancement</u> (in offspring) ; 6 <u>F₁</u> have, same genotype / uniform phenotype ; 7 AVP ;	4
6(b)(i)	any one from: 1 (IR8) less likely to, fall over / lodge / collapse ; 2 (IR8) higher proportion / more, of plant is, edible / grain ;	1
6(b)(ii)	check for, genes / alleles / DNA, for high yield and flood tolerance ;	1
6(b)(iii)	any one from: 1 to, maintain high yield / increase yield / give higher yield ; 2 to avoid inbreeding depression or to maintain, hybrid vigour / heterosis ;	1

Question	Answer	Marks
6(c)(ii)	any four from: variety A: 1 gibberellin binds to, receptor / GID1 ; 2 DELLA protein breaks down ; 3 PIF / transcription factor (released) to, bind to DNA / transcribe genes ; 4 stem / internode / cell, elongation / growth, occurs ; 5 A, is <i>lele</i> / has no <i>Le</i> allele ; 6 A does not produce its own (active) gibberellin ; variety B: 7 does not respond to / no effect from, gibberellin ; 8 lacks / has non-functional, gibberellin receptor ;	4
6(c)(ii)	any two from: 1 (BZR1) binds to <u>promoter</u> ; 2 to switch gene on / so gene is expressed / allow transcription ; 3 allows binding of <u>RNA polymerase</u> (to DNA / promoter) ;	2